

Executive Presentation:

Reservoir Stage Kinematics & Modeling

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Document Reference: lab02_villondo.html

Course: Numerical Methods in Civil Engineering

Dataset Analyzed: Data-01.xlsx

Date: August 25, 2026

Core Methodology: Finite-Difference Derivatives, 4-Parameter Logistic Fitting, & Numerical Integration

1. Executive Summary

- **Primary Objective:** Analyze stage height $h(t)$ over time, determine peak inflow rates, fit a non-linear hydrodynamic model, and verify total volumetric stage accumulation.
- **Key Findings:**
 - **Model Performance:** High precision fit with a coefficient of determination $R^2 = 0.99982$ and residual standard error $s = 0.0412$ m.
 - **Hydrodynamic Peak:** Maximum inflow velocity achieved at $t = 14.82$ hours with a rate of $\frac{dh}{dt} = 0.4812$ m/h.
 - **Volumetric Consistency:** Continuous numerical integration (quad) and discrete trapezoidal integration (trapz) matched with less than **0.02% variance**.

2. Kinematics & Rate Analysis

Derivative Calculations ($\frac{dh}{dt}$ & $\frac{d^2h}{dt^2}$)

- Methodology:** Central finite differences applied to interior nodes with 2nd-order single-sided formulas at endpoints.
- Key Metrics:**
 - Maximum Inflow Rate (Max $\frac{dh}{dt}$):** 0.4812 m/h
 - Peak Hour (t_0):** 14.82 hours

Hydrodynamic Observation: The 2nd derivative ($\frac{d^2h}{dt^2}$) indicates that peak stage acceleration occurs immediately prior to $t = 14.82$ hours, after which deceleration begins as the reservoir approaches storage capacity.

3. 4-Parameter Logistic Model Fitting

Mathematical Model

$$h(t) = c + \frac{a}{1 + e^{-k(t-t_0)}}$$

Fitted Parameter Specifications

Parameter	Description	Estimated Value	Standard Error	Statistical Significance
c	Base Stage Level	2.1042 m	0.0084	$p < 0.0001$ (Significant)
a	Stage Capacity Range	8.4110 m	0.0121	$p < 0.0001$ (Significant)
k	Growth Rate Constant	0.2289 h ⁻¹	0.0018	$p < 0.0001$ (Significant)

Parameter	Description	Estimated Value	Standard Error	Statistical Significance
t_0	Inflection Point Hour	14.8211 h	0.0415	$p < 0.0001$ (Significant)

- **Conclusion:** All parameters show high statistical significance ($p \ll 0.05$), confirming the 4-parameter logistic curve accurately models physical storage constraints.

4. Model Diagnostics & Error Analysis

Goodness-of-Fit Diagnostics

- **Sum of Squared Errors (SSE):** 0.084891
- **Total Sum of Squares (SST):** 471.5812
- **Coefficient of Determination (R^2):** 0.999820
- **Residual Standard Error (s):** 0.041203 m

Residual Behavior

- Residuals are symmetrically distributed around zero with no systematic bias or spatial drift.
- Maximum deviation between empirical observations and fitted values remains within ± 0.08 meters.

5. Volumetric Integration & Cross-Verification

Stage-Time Area Integration Comparison ($\int h(t) dt$)

Integration Method	Computed Value	Difference
Adaptive Quadrature (scipy.integrate.quad)	184.6214 m · h	Baseline

Integration Method	Computed Value	Difference
Trapezoidal Rule (np.trapezoid)	184.6512 m · h	+0.0298 m · h
Percentage Variance	0.0161%	High Concordance

- **Final Note:** The 0.0161% variance confirms robust numerical stability between continuous model integration and discrete empirical calculations.